# 1. Have you used AI before?

Prior to enrolling in this course, I have used and engaged with various Artificial Intelligence (AI) tools in both my studies and daily life:

### AI tools I have used:

- **Large Language Models (LLMs):** ChatGPT (OpenAI), Google Gemini, Claude (Anthropic).
- **AI Coding Assistants:** GitHub Copilot.
- **Study & research tools:** Notion AI.

### Specific contexts and purposes:

1. **Concept explanation & academic support:**
    - Used AI as a virtual tutor to explain abstract concepts (e.g., data structures, Big-O algorithm complexity, Git workflow) using simple language and visual examples.
    - Assisted in reading, understanding, and quickly summarizing technical documentation and research papers in English.
2. **Coding support & debugging:**
    - Pasted compiler error messages and stack traces into AI to get suggestions on root causes and solutions.
    - Asked AI to suggest test cases for edge cases that I might have overlooked.
3. **Productivity improvement:**
    - Outlined articles, corrected grammar, refined academic writing style, and optimized study schedules.

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# 2. Do you have any experience or tips for using AI?

Through hands-on experience using AI, I have learned several core principles and tips to maximize its potential without losing independent thinking:

### Tip 1: Precise and context-rich prompt engineering

- **The C-R-I-O framework:** When giving tasks to AI, provide four elements:
    - **C (Context):** Who you are, what problem you are solving, and what language/technology you are using.
    - **R (Role):** Give the AI a role (e.g., "*Act as a veteran software engineer specializing in C++*").
    - **I (Instruction):** Provide concrete instructions, clearly stating what to do and what not to do.
    - **O (Output format):** Specify the desired format (tables, Markdown, or concise bullet points).
- **Break down problems:** Avoid vague, monolithic requests. Instead, break problems into logical steps and prompt the AI iteratively (Chain-of-thought prompting).

### Tip 2: "Verify, Don't Trust Blindly"

- AI always carries the risk of **"hallucination"**—generating inaccurate information, non-existent libraries, or made-up functions convincingly.

- **Mandatory rule:** Always cross-check AI answers with official documentation, and manually run and verify every line of code before using it in a project.

### Tip 3: Treat AI as a "Co-pilot", not a "Replacement"

- Do not use AI to do all the coursework without understanding the core concepts.
- Use AI to stimulate your thinking. For instance, instead of asking "*Solve this problem for me*", ask "*Give me 3 approaches to optimize this problem*", or "*Why is my code running with $O(n^2)$ complexity instead of $O(n)$?*".
- You must be the one who thoroughly understands the final solution to confidently present and answer questions from instructors.

### Tip 4: Data privacy & academic integrity

- **Never share sensitive data:** Avoid entering personal private information, passwords, API keys, or proprietary source code onto public AI platforms.
- **Transparency & accountability:** Always be honest and declare your AI usage (AI Usage Declaration) according to course policies, and take full responsibility for the accuracy and safety of the output.

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## 3. Personal Takeaway

AI is a breakthrough for humanity, providing a powerful tool to boost study and research efficiency. However, AI only brings real value when driven by sharp critical thinking and academic ethics. As an IT student, I commit to using AI responsibly, proactively improving my skills, and strictly following the course regulations.